

Basic Principles of Medicine 1

Module: CPR Module DLA

Topic:

Local Control of Blood Flow

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2. PHYS.0121	Define autoregulation of blood flow. Distinguish between short-term and long-term autoregulatory responses and the mechanisms responsible for each.
SOM.MK.II.BPM1.3.CPR.2. PHYS.0122	Describe how the theory of metabolic regulation of blood flow accounts for active hyperemia and reactive hyperemia.
SOM.MK.II.BPM1.3.CPR.2. PHYS.0123	Describe the contribution of myogenic tone to blood flow regulation.
SOM.MK.II.BPM1.3.CPR.2. PHYS.0124	Identify the role of PO ₂ , PCO ₂ , pH, adenosine, and K ⁺ in the metabolic control of blood flow to specific tissues.
SOM.MK.II.BPM1.3.CPR.2. PHYS.0125	Illustrate and explain the synthetic pathway for nitric oxide including substrate and the interplay between endothelium and vascular smooth muscle
SOM.MK.II.BPM1.3.CPR.2. PHYS.0126	Discuss the circumstances and the mechanisms whereby humoral substances contribute to regulation of the microcirculation.
SOM.MK.II.BPM1.3.CPR.2. PHYS.0127	Discuss the interaction of a) intrinsic (local), b) neural, and c) humoral control mechanisms and contrast their relative dominance in the CNS, coronary, cutaneous, and skeletal muscle vascular beds.

Recommended Reading

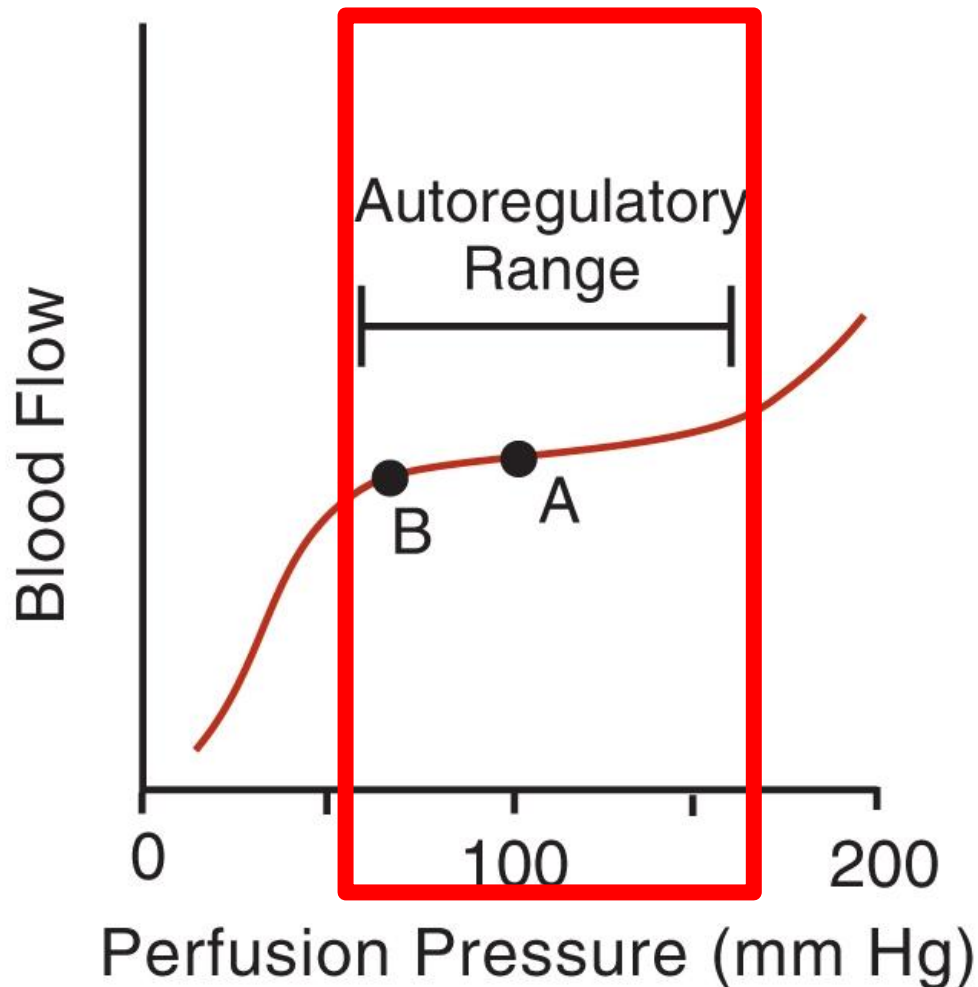
1. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell. Chapter 15: Regulation of Organ Blood Flow and Capillary Transport
2. Cardiovascular Physiology Concepts, 3e. Richard E. Klabunde. Chapter 7: Organ Blood Flow- Read under heading, “Local control of Blood flow”.

AUTOREGULATION OF BLOOD FLOW

Autoregulation of Blood Flow in an organ

- Definition: Autoregulation is the intrinsic ability of an organ to maintain a **constant blood flow** despite changes in certain range of perfusion pressure.
- If blood pressure to an organ falls, the blood flow will initially fall, then return toward normal levels over the next few minutes.
- If the blood pressure increases, the blood flow initially increase, then return toward normal over the next few minutes.
- This autoregulatory response is a local or **intrinsic response of the organ** and **NOT dependent on extrinsic factors like nerves or hormones**.
- The range of perfusion pressure within which autoregulation occurs is called **autoregulatory range** of perfusion pressure.

Autoregulation of Blood Flow in an organ



In this figure, if the perfusion pressure from point A changes to point B, the blood flow to the organ remains the same.

Autoregulation maintains a constant blood flow in the pressure range between 60 to 160 mm Hg (red box).

The organ fails to maintain a constant blood flow below or above this autoregulatory range of perfusion pressure.

Ability to autoregulation of Blood Flow varies in different organs

- **Most efficient** autoregulation occurs in coronary (**heart**), cerebral (**brain**) and renal (**kidney**) circulations.
- Moderate autoregulation occurs in skeletal muscles and gastrointestinal system.
- There is **no autoregulation** in skin (cutaneous circulation).
- Importance: Autoregulation protects vital organs like brain and heart in conditions of low or high blood pressures.

Examples: In condition of stenosis of coronary vessels, autoregulation maintains normal blood flow and normal delivery of oxygen to heart muscles.

In conditions of low blood pressure due to blood loss, the blood flow to brain and heart may not be compromised.

The mechanisms of autoregulation of Blood Flow

- Two mechanisms explain autoregulation in organs:
 1. Myogenic mechanism
 2. Metabolic mechanism.
- **Myogenic mechanism:** An increase in blood pressure stretches the blood vessels. Smooth muscles of the arterioles contract in response to this stretch. Vasoconstriction by smooth muscle contraction reduces the radius of the blood vessels and reduces blood flow. Opposite occurs when the blood pressure falls.
- **Metabolic mechanism:** If the perfusion pressure to an organ is reduced, there would be lack of oxygen and accumulation of metabolites. These factors relax smooth muscles and dilate the blood vessels resulting in increase in blood flow.

Explanation for Myogenic mechanism of autoregulation

Increase in blood flow to an organ by a rise in the perfusion pressure



Stretching of the arterioles in the organ



Stimulation of **stretch-activated calcium channels** in vascular smooth muscles of the arterioles



Smooth muscle contraction (**vasoconstriction**) of the arterioles in the organ



Blood flow reduces and comes back to normal amount

Decrease in blood flow to an organ by a fall in the perfusion pressure



Decreased stretching of the arterioles in the organ



Inhibition of stretch-activated calcium channels in vascular smooth muscles of the arterioles



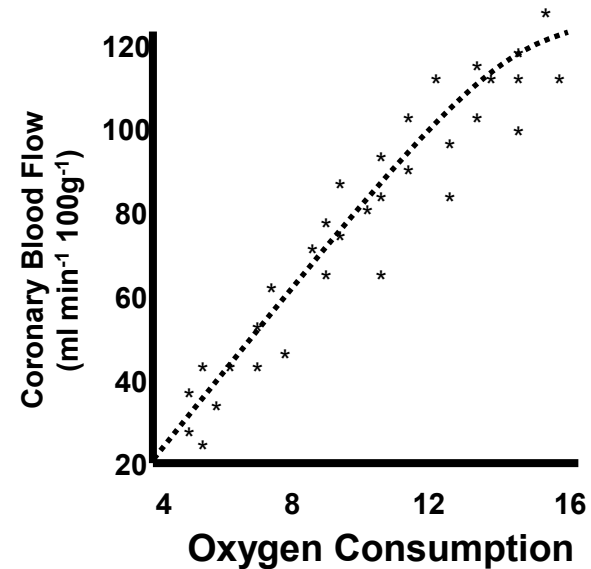
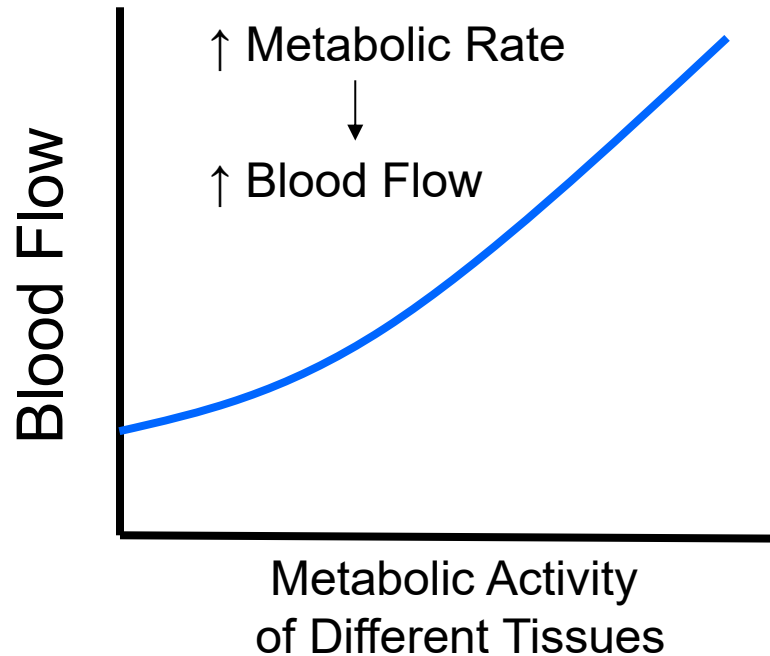
Smooth muscle relaxation (**vasodilation**) of the arterioles in the organ



Blood flow improves and comes back to normal amount

METABOLIC REGULATION OF BLOOD FLOW TO AN ORGAN

Tissue Metabolic Need Determines Blood Flow to an organ



Oxygen consumption of a tissue increases with metabolic activity. Increased metabolic rate correlates well with increased blood flow.

Examples: Blood flow in i) coronary vessels (figure on the right) and ii) in skeletal muscle increase during exercise.

Metabolic Regulation of Blood Flow to an organ

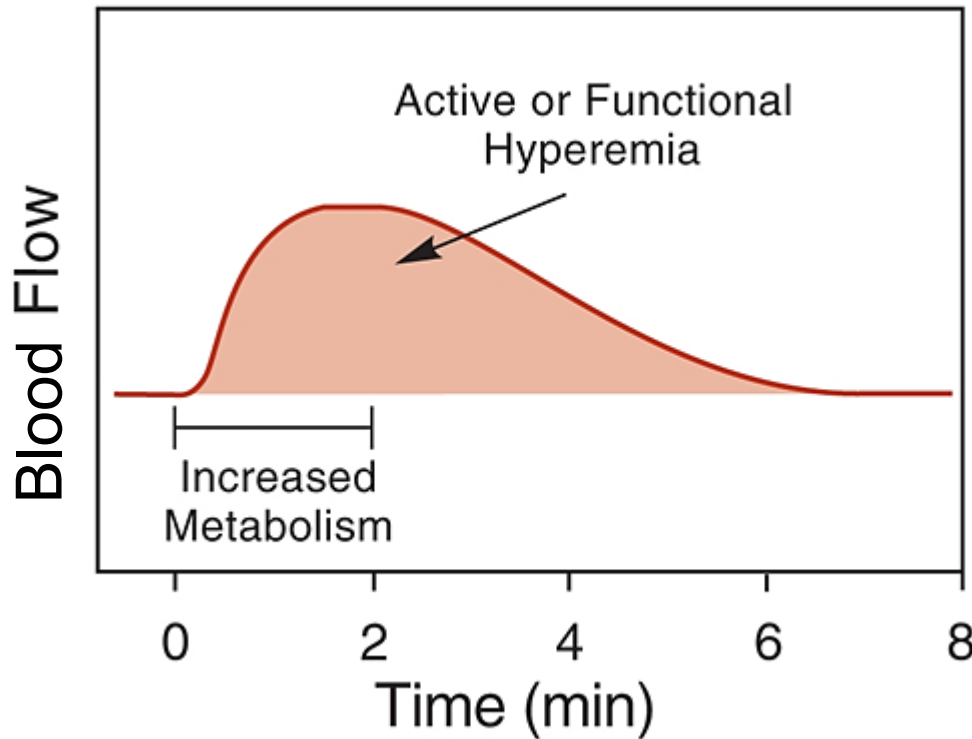
- Metabolic activity in the tissue regulates its blood flow locally without the influence of nerves or hormones.
- Increase in blood flow to an organ is called hyperemia that consists of 2 types, namely, **Active hyperemia** and **Reactive hyperemia**.

Active hyperemia: An increase in blood flow to an organ during an increased metabolic activity of an organ or tissue. It is also known as functional hyperemia.

Examples: Active hyperemia occurs in skeletal muscles and heart during exercise; Increased blood flow in gastrointestinal tract during digestion and absorption of food.

The graphs in the next 2 slides explain this hyperemia.

ACTIVE HYPEREMIA

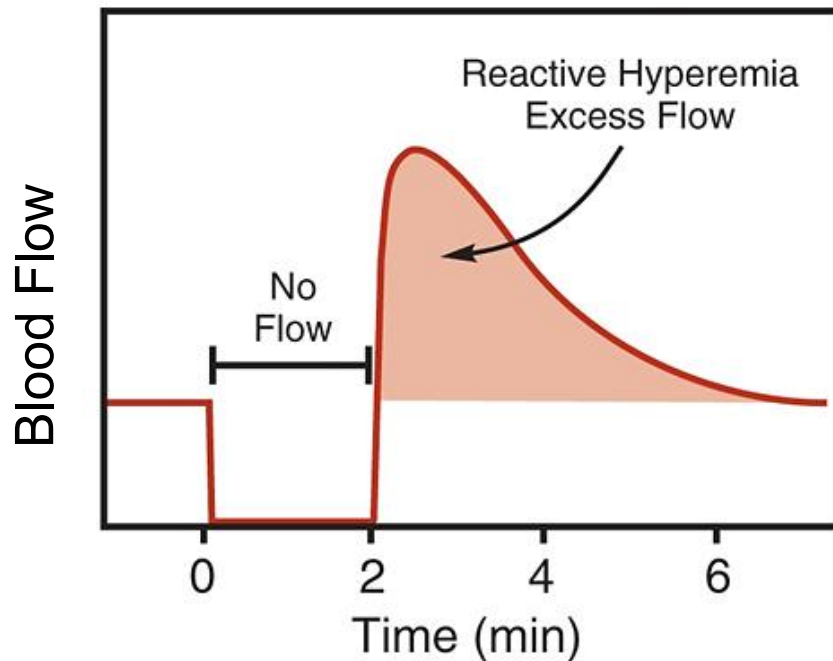


Increase in Blood Flow to an organ occurs because of an increased production of metabolites and endothelial factors that cause vasodilation in the organ.

This explains the increase in blood flow to skeletal and cardiac muscles during exercise.

REACTIVE HYPEREMIA

Reactive hyperemia is the transient increase in organ blood flow that occurs **following a brief period of stoppage** of blood flow. It is also known as post-ischemic hyperemia.



The longer the period of occlusion, longer will be the subsequent reactive hyperemia.

Mechanism:

Temporary occlusion of a blood vessel



Build-up of metabolites downstream of the occlusion



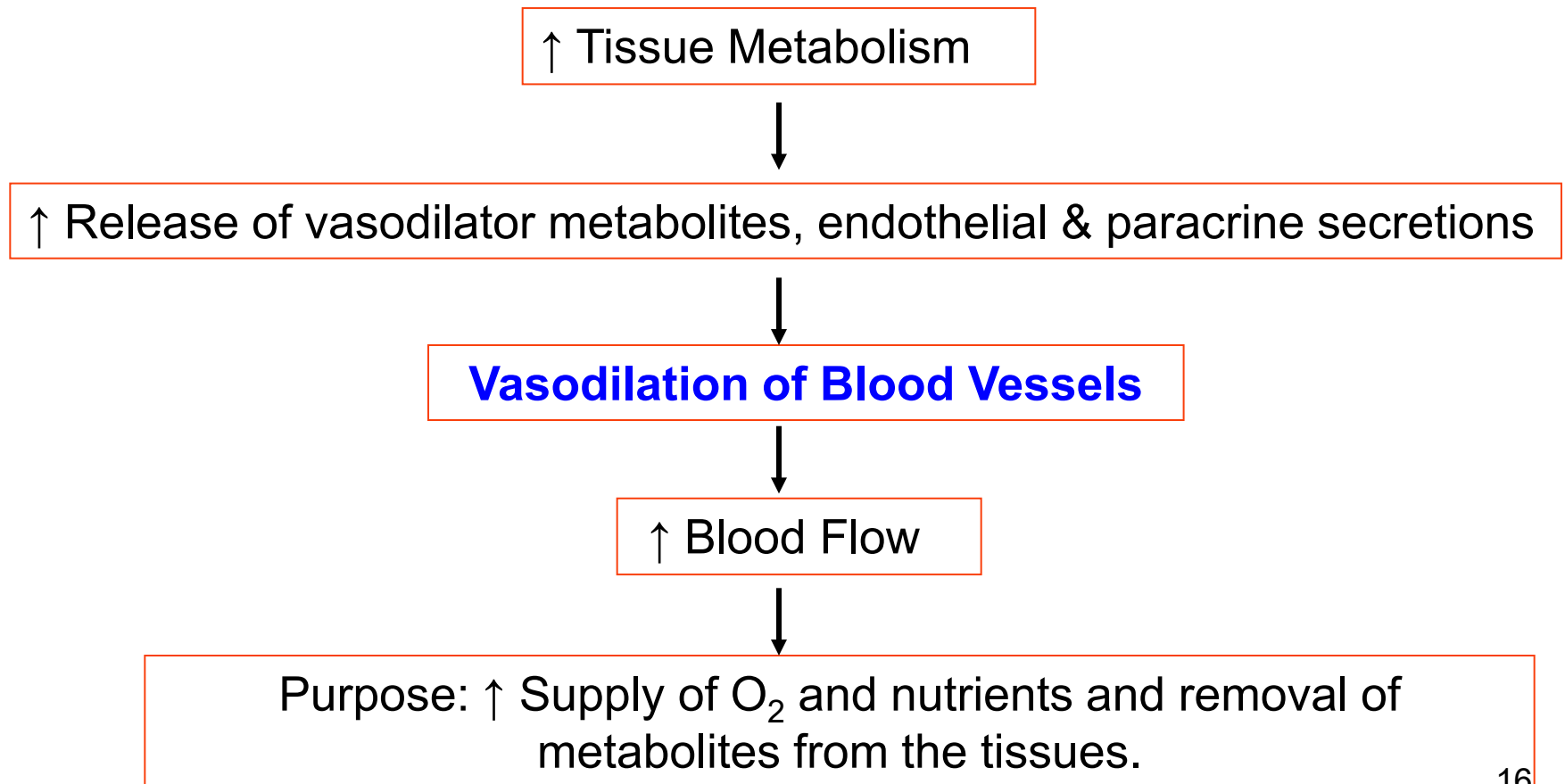
Vasodilation downstream



Excess Blood Flow

METABOLIC CONTROL OF BLOOD FLOW

In both active and reactive hyperemia, the metabolites in the tissues increase blood flow.



MAJOR VASODILATORS IN TISSUES THAT CONTRIBUTE LOCAL CONTROL OF BLOOD FLOW

MAJOR VASODILATORS in Tissues

The most common vasodilators in the tissues include:

- Adenosine
- Increased CO₂
- Increased H⁺ ion and lactic acid
- Low Oxygen
- Potassium.

Some of these vasodilators play major role in vasodilation in certain tissues as explained in the next slide.

TISSUE VASODILATORS

Adenosine: Formed from AMP (adenosine monophosphate) and a vasodilator in most of the tissues, especially in the **coronary circulation**. However, adenosine causes vasoconstriction in kidney (exception).

CO₂ & H⁺: Cerebral blood vessels are very sensitive to increased CO₂ and [H⁺]. Elevated arterial PCO₂ is an important regulator of **cerebral circulation**.

Low PaO₂: Low O₂ (hypoxia) causes vasodilation either directly or indirectly (by releasing adenosine, lactic acid or H⁺) in almost all tissues EXCEPT lungs. **Exception: hypoxia constricts pulmonary blood vessels.**

Interstitial K⁺: Increased skeletal muscle activity increases action potentials which causes K⁺ efflux and thus, increases interstitial K⁺. Increase in **skeletal muscle blood flow** during exercise is mainly by this potassium.

Explain: How hyperkalemia produces vasodilation in vascular smooth muscle:

In all normal cells:

Increase in K^+ on the outside of the cell decreases the driving force and causes the resting membrane potential to become more positive. This may depolarize the cell quicker.

In vascular smooth muscle:

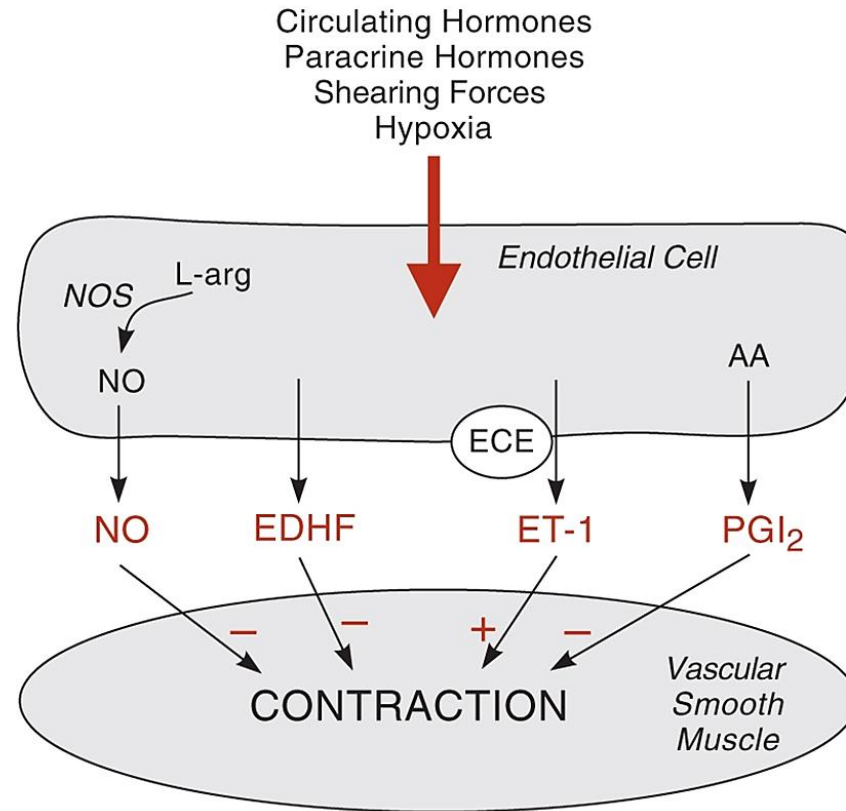
Increase in K^+ on the outside of the cell increases K^+ conductance through special potassium channels in smooth muscle cells. This results in K^+ moving out of the cell and causes **hyperpolarization**. **Depolarization does not occur in this tissue!** Hyperpolarization decreases intracellular calcium and causes smooth muscle relaxation and vasodilation.

ROLE OF ENDOTHELIAL FACTORS IN VASODILATION OR VASOCONSTRICTION

- Vascular endothelium releases certain substances that could cause smooth muscle relaxation or contraction and thus can control blood flow to an organ/tissue.
- Endothelial cells release these substances when the cells are exposed to **several influences like hormones, mechanical stress, hypoxia or drugs.**
- The major endothelial-derived vasoactive substances include:
 - Nitric oxide (NO) - vasodilator
 - Prostacyclin* (PGI₂) - vasodilator
 - Endothelin-1 (ET-1) – vasoconstrictor.

*Prostacyclin is also known as prostaglandin-I₂ (PGI₂)

ENDOTHELIAL-DERIVED VASOACTIVE SUBSTANCES



Note:

- Nitric oxide (NO) is formed by nitric oxide synthase (NOS) using l-arginine (l-arg).
- Endothelial-derived hyperpolarizing factors (EDHF)-less important substance.
- Negative sign (-) indicates relaxation (vasodilation) and positive (+) indicates vasoconstriction of the vascular smooth muscle.

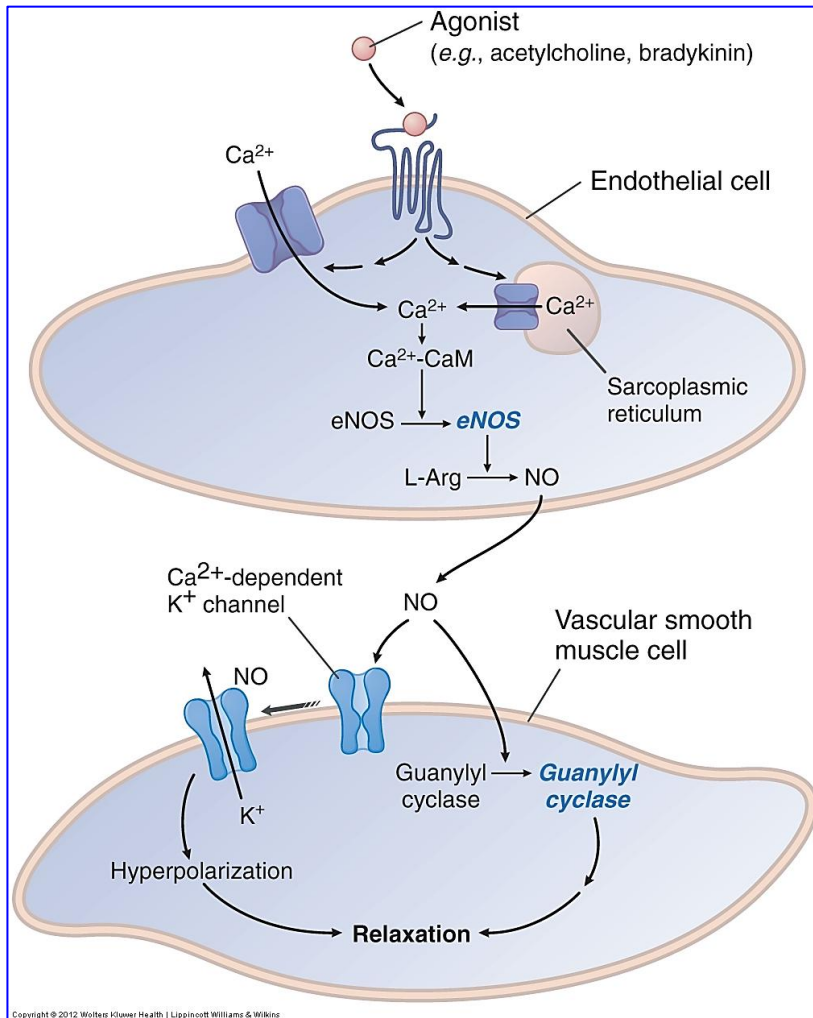
ENDOTHELIAL-DERIVED VASOACTIVE SUBSTANCE: Nitric oxide (NO)

NITRIC OXIDE (NO): a vasodilator

- Main stimulus for NO production is shear stress at endothelial cell.
- Other factors such as acetylcholine, histamine, bradykinin, and hypoxia also stimulate release of NO.
- NO diffuses to vascular smooth muscle cells and stimulates guanylyl cyclase $\rightarrow \uparrow$ cGMP $\rightarrow$ relaxation of smooth muscle $\rightarrow$ Vasodilation.
- NO mediates numerous **beneficial effects in the body** such as:
 - Responsible for flow-induced vasodilation in cerebral, skeletal and intestinal vasculature,
 - Responsible for vasodilation in erectile tissue and erection during sex,
 - Offers tonic anti-hypertensive effect,
 - Produces coronary vasodilation. Nitrates used as drugs in angina release NO and relieve anginal pain by improving circulation.

Note: Too much NO production during septicemia due to bacterial infection may cause vasodilation, distributive shock and death!

Nitric Oxide (NO) causes Smooth Muscle Relaxation by its paracrine action



Paracrine activity of NO:

↓

Agonists like Ach & bradykinin bind to the endothelial cells

↓

Increased enzyme, **NO synthase (eNOS)** activity to produce NO in endothelial cells

↓

NO diffuses to vascular smooth muscle cells

↓

Stimulation of Guanylyl cyclase and increased production of **cGMP**

↓

Smooth muscle relaxation and Vasodilation

ENDOTHELIAL-DERIVED VASOACTIVE SUBSTANCE: Endothelin-1 (ET-1)

Endothelin-1 (ET-1): a Vasoconstrictor

- Main stimuli for ET-1 release are angiotensin II, vasopressin and inhibited by nitric oxide.
- ET-1 binds to ET_A receptors on smooth muscle cells, which are coupled to Gq-proteins → Increases IP3 and calcium → Smooth muscle contraction Vasoconstriction.
- Increased ET-1 mediates **many harmful effects in the body** such as:
 - Causes vasoconstriction in lungs and produces pulmonary hypertension.
 - Causes vasoconstriction in renal and peripheral blood vessels that leads to renal failure, hypertension and heart failure.
 - Cause spasmodic contractions in cerebral blood vessels that leads to brain injury and stroke.

IMPORTANCE OF ENDOTHELIAL-DERIVED VASOACTIVE SUBSTANCES: Clinical Correlates

- **Endothelial dysfunction** occurs in major diseases like arteriosclerosis, hypertension, diabetes mellitus and hypercholesterolemia.
- Endothelial damage and dysfunction results in less NO and PGI₂ (vasodilators) production and increased ET-1 (vasoconstrictor) production.
- Loss of vasodilatory capacity and increased vasoconstriction further aggravates the above conditions, leading to complications such as heart failure, pulmonary hypertension and renal failure.

NEURAL CONTROL OF BLOOD FLOW TO AN ORGAN

Neural control: Role of sympathetic nervous system

- Sympathetic nervous system (SNS) innervates arterioles and veins in all organs *except brain*.
- Stimulation of these nerves release norepinephrine as neurotransmitter.
- **All organs** except brain and heart have both alpha-1 (vasoconstriction) and beta-2 (vasodilation) receptors. As alpha receptors exceed beta in these tissues, the overall effect is **vasoconstriction**. Removal of sympathetic tone cause vasodilation in these tissues.
- Heart blood vessels contain only **beta-2 receptors**.
- SNS can change organ blood flow, but *it is not used to control local organ blood flow*.
- SNS contributes to the total peripheral resistance in the body to control the systemic arterial **blood pressure of the whole body**.

Role of sympathetic nervous system (continued)

- SNS can take the control of local blood flow in situations that demands control of **whole-body blood pressure**.
- If the systemic blood pressure drops significantly (example: in bleeding), stimulation of the **SNS will override local regulatory mechanisms in the tissues** to control the whole-body blood pressure. Under this condition, blood flow to all organs except heart and brain may be compromised.
- During exercise, stimulation of the SNS decreases blood flow to almost all organs (except brain and heart), and favors blood flow to the skeletal muscles. This differential effect helps to maintain whole-body blood pressure.
- ***Neural control is mainly concerned with control of mean arterial pressure (MAP) and total peripheral resistance.***

Practice Question 1

A 58-year-old man is posted for a renal transplant because of end stage kidney disease. The surgeon removes the kidney of the donor by clamping the blood vessels. This kidney is immediately transplanted to the recipient by surgery. After successful transplantation to the patient, the surgeon removes the clamps to allow the blood flow to the transplanted kidney. It is noticed that the blood flow to this kidney increases for a few minutes and then comes back to a normal level. Which of the following mechanisms best explains the initial rise in renal blood flow in this patient?

- A. Active hyperemia
- B. Reactive hyperemia
- C. Hypoxic vasoconstriction
- D. Release of vasoconstrictors
- E. Sympathetic nerve stimulation.

Correct Answer: B. Refer to slide#15. Temporary occlusion causes reactive hyperemia in the donor kidney.

Practice Question 2

A 64-year-old woman comes to the physician because of a 4-month history of progressive shortness of breath, swollen ankles, and fatigue. She has a 6-year history of uncontrolled hypertension. Physical examination shows pitting edema of the lower limbs and bilateral basal crepitations in lung fields. Cardiovascular studies show an elevated total peripheral resistance due to the release of vasoconstrictors in the peripheral tissues. Which of the following substances is most likely released in this patient?

- A. Nitric oxide
- B. Prostacyclin (PGI_2)
- C. Lactic acid
- D. Endothelin-1 (ET-1)
- E. Adenosine

Correct Answer: D. Refer to slides 21 & 25. All other choices are vasodilators in peripheral tissues.

Any Questions ?

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